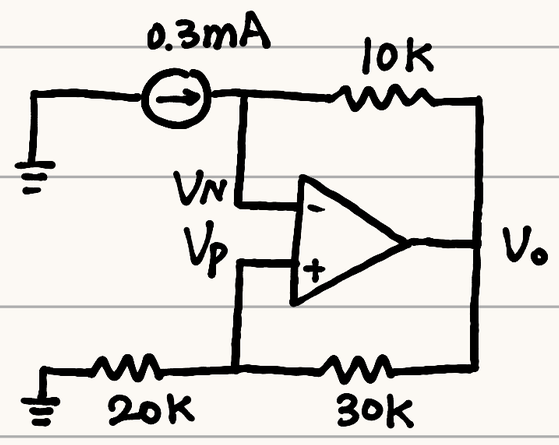
1.18

(a)

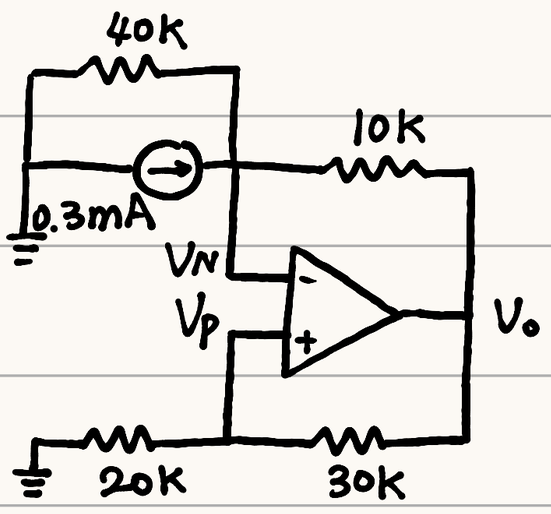


$V_o \times \frac{20k}{20k+30k} = V_N = V_p \Rightarrow 0.4V_o = V_N = V_p$

$V_N - 10k \cdot 0.3mA = V_o \Rightarrow V_N = V_o + 3$

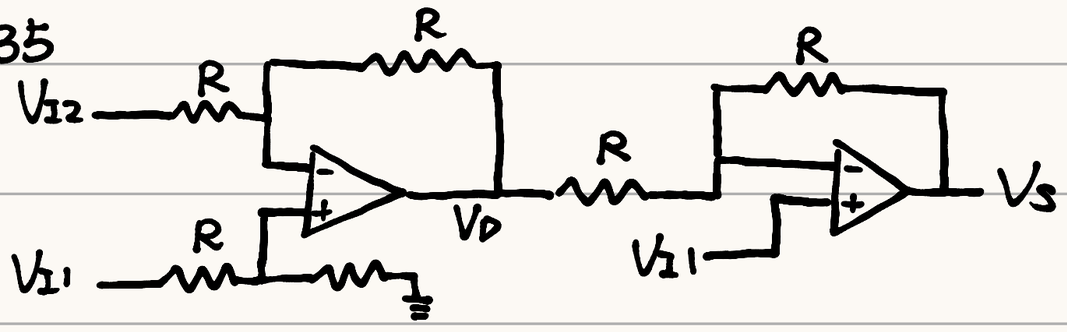
$\Rightarrow 0.4V_o = V_o + 3 \Rightarrow V_o = -5V \quad V_N = V_p = -2V$

(b)

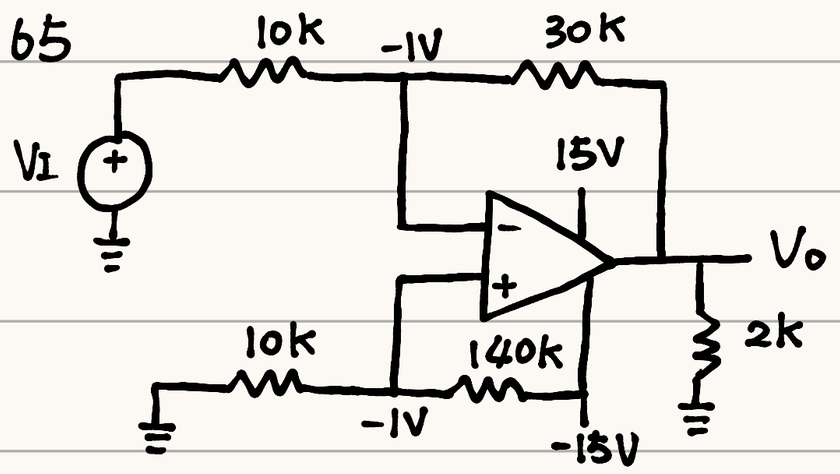


$$\begin{cases} \frac{V_N - V_o}{10k} + \frac{V_N}{40k} = 0.3mA \\ 0.4V_o = V_N = V_p \end{cases} \Rightarrow \begin{cases} V_o = -6V \\ V_N = V_p = -2.4V \end{cases}$$

1.35



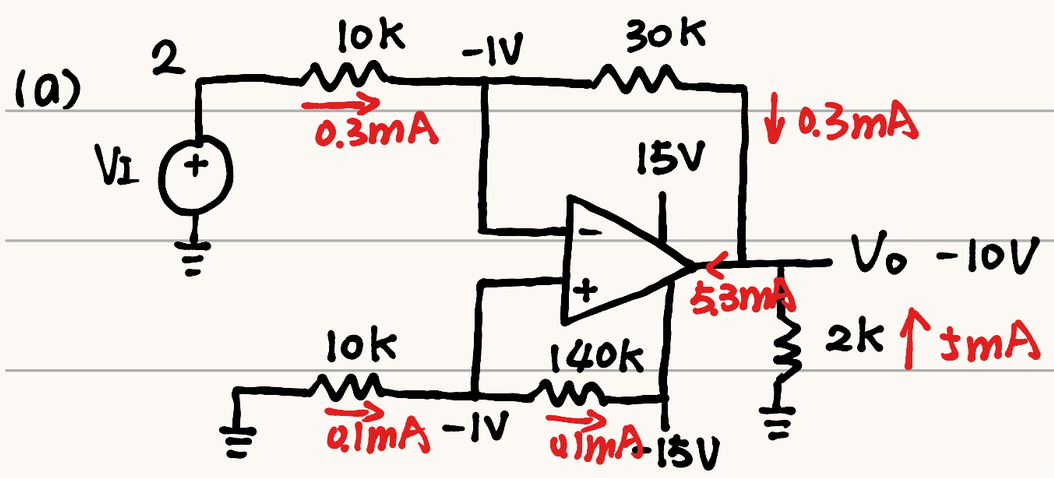
1.65



$\frac{V_I + 1}{10k} = \frac{-1 - V_o}{30k} \Rightarrow V_o = -4 - 3V_I$

2

(a)

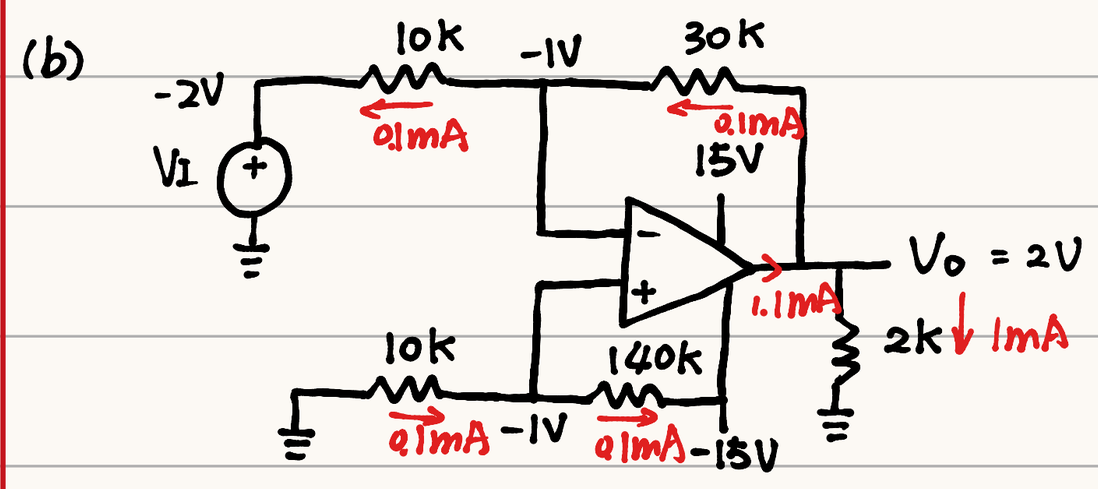


$V = -10 - (-15) = 5V$

$P = U \cdot I = 5 \times 5.3mA = 26.5mW$

$P_Q = 30V \times 1.5mA = 45mW \quad P_t = 71.5mW$

(b)



$U = 15 - 2 = 13V \quad P = U \cdot I = 13 \times 1.1mA = 143mW$

$P_t = 45mW + 143mW = 59.3mW$